

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION

To overcome the limitations of traditional comic creation and standalone AI image generators, we propose **ComicCraft**, a comprehensive web-based platform. ComicCraft acts as an automated virtual comic studio, seamlessly merging Large Language Models (LLMs) with advanced text-to-image Diffusion models to generate cohesive comic stories.

3. OVERVIEW OF THE PROPOSED SYSTEM:

The proposed system is built on the FastAPI framework to ensure rapid backend processing. It provides a clean HTML/CSS frontend where users can define their creative vision via input fields (Prompt, Character, Setting, Tone, Art Style). Upon submission, the system executes a sequential pipeline: structuring the story, generating the narrative, generating the images, rendering a web preview, and exporting a PDF.

4. KEY FEATURES OF THE SOLUTION

- **Intuitive User Interface:** A simple Jinja2-powered HTML form that abstracts all complex AI interactions away from the user.
- **Dual-Layered Text Generation:** Uses two different Gemini models optimized for specific tasks (Flash for JSON structuring, Pro for creative storytelling).
- **Automated Image Prompting:** The system internally translates the user's basic story idea into highly detailed engineering prompts required by Stable Diffusion, ensuring high-quality outputs.
- **Dynamic Web Preview:** An interactive `comic_preview.html` page that displays the generated comic sequentially, panel by panel, complete with titles and narrations.

- **One-Click PDF Export:** A fully automated backend process that binds the text and images into an offline, shareable PDF format using the FPDF library.

5. TECHNICAL APPROACH

- **Google Gemini 1.5 Flash:** Selected for its speed and accuracy in generating strict JSON arrays. It is tasked with creating the 5-panel layout outline.
- **Google Gemini 1.5 Pro:** Selected for its advanced natural language processing. It takes the outline and acts as a comic book writer to produce engaging dialogue and narration.
- **Hugging Face Diffusers (Stable Diffusion v1.5):** Chosen for its superior ability to generate vibrant, comic-style illustrations dynamically from the AI-generated scene descriptions.

6. USER WORKFLOW

1. The user visits the homepage and fills out the comic details.
2. The user submits the form, initiating the AI backend pipeline.
3. The system presents the complete comic on a preview page.
4. The user clicks "Download Your Comic as PDF" and receives a formatted document.

7. ADVANTAGES OF THE PROPOSED SOLUTION

The proposed solution eliminates the need for manual design, minimizes user effort, drastically reduces creation time from weeks to seconds, and provides a highly scalable architecture capable of supporting diverse art styles and storytelling genres.